

Cyclotomic multiple zeta values of level 2^n or 3^n

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Today's talk

- Introduction and main theorem
- Sketch of proof for weight 2 case
- Some observations for other levels, weight 2
- Some observations for integral structure

Introduction and main theorem

Cyclotomic Multiple Zeta Values (1/2)

Let $\mu_N := \{\zeta_N^a \mid a \in \mathbb{Z}/N\mathbb{Z}\}$ be the set of N -th roots of unity.

Definition (Cyclotomic Multiple Zeta Values (=CMZVs))

For $k_1, \dots, k_d \in \mathbb{Z}_{\geq 1}$ and $\epsilon_1, \dots, \epsilon_d \in \mu_N$ such that $(k_d, \epsilon_d) \neq (1, 1)$, we define

$$\zeta \left(\begin{matrix} k_1, & \dots, & k_d \\ \epsilon_1, & \dots, & \epsilon_d \end{matrix} \right) := \sum_{0 < m_1 < \dots < m_d} \frac{\epsilon_1^{m_1} \dots \epsilon_d^{m_d}}{m_1^{k_1} \dots m_d^{k_d}}.$$

$$* \quad \zeta \left(\begin{matrix} k_1, & \dots, & k_d \\ 1, & \dots, & 1 \end{matrix} \right) = \sum_{0 < m_1 < \dots < m_d} \frac{1}{m_1^{k_1} \dots m_d^{k_d}} := \zeta(k_1, \dots, k_d)$$

* N : **level.** $k_1 + \dots + k_d$: **weight.** d : **depth.**

Dimension of CMZVs

A main interest is **linear relations** among CMZVs.

The basic quantity is a **dimension**.

It is strongly believed that there are no linear relations among CMZVs of different weight.

Let $Z_w^{(N)} := \langle \text{cyclotomic MZVs of level } N \text{ and weight } w \rangle_{\mathbb{Q}} \subset \mathbb{C}$.

What is $\dim_{\mathbb{Q}} Z_w^{(N)}$?

Example (level 2, weight 2)

$$\zeta \left(\begin{smallmatrix} 1, & 1 \\ +1, & -1 \end{smallmatrix} \right) = \sum_{0 < m < n} \frac{(-1)^n}{mn} = \frac{1}{2} \log(2)^2, \quad ,$$

$$\zeta \left(\begin{smallmatrix} 2 \\ +1 \end{smallmatrix} \right) = \sum_{0 < n} \frac{1}{n^2} = \frac{1}{6} \pi^2,$$

$$\zeta \left(\begin{smallmatrix} 1, & 1 \\ -1, & -1 \end{smallmatrix} \right) = \sum_{0 < m < n} \frac{(-1)^{m+n}}{mn} = -\frac{1}{12} \pi^2 + \frac{1}{2} \log(2)^2, \quad \zeta \left(\begin{smallmatrix} 2 \\ -1 \end{smallmatrix} \right) = \sum_{0 < n} \frac{(-1)^n}{n^2} = -\frac{1}{12} \pi^2.$$

$$\Rightarrow Z_2^{(2)} = \mathbb{Q} \log(2)^2 + \mathbb{Q} \pi^2$$

By Baker's theorem, π and $\log(2)$ are linearly independent over $\bar{\mathbb{Q}}$

$$\Rightarrow \dim_{\mathbb{Q}} Z_2^{(2)} = 2$$

Example (level 1, weight 5)

$$\zeta(1,1,1,2) = \zeta(5), \quad \zeta(1,4) = \zeta(1,1,3) = 2\zeta(5) - \zeta(2)\zeta(3)$$

$$\zeta(2,3) = \zeta(1,2,2) = -\frac{11}{2}\zeta(5) + 3\zeta(2)\zeta(3), \quad \zeta(3,2) = \zeta(2,1,2) = \frac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3)$$

$$\Rightarrow Z_5^{(1)} = \mathbb{Q}\zeta(5) + \mathbb{Q}\zeta(2)\zeta(3)$$

Thus,

$$\dim_{\mathbb{Q}} Z_5^{(1)} = \begin{cases} 2 & \text{if } \zeta(5) \text{ and } \zeta(2)\zeta(3) \text{ are linearly independent} \\ 1 & \text{if } \zeta(5) \text{ and } \zeta(2)\zeta(3) \text{ are linearly dependent} \end{cases}$$

Linear independence of $\zeta(5)$ and $\zeta(2)\zeta(3)$ are not known and seems quite difficult.

Why do we consider Motivic CMZVs ?

- ✱ To show $\dim_{\mathbb{Q}} Z_5^{(1)} = 2$, we need linear independence of $\zeta(5)$ and $\zeta(2)\zeta(3)$.
- ✱ In fact, for most cases, the proof of such linear independence is too difficult.
- ✱ Thus, we consider the dimension of **motivic CMZVs** instead for the one of CMZVs.

Motivic CMZVs are lifts of CMZVs.

$$\begin{array}{ccc}
 \mathcal{H}^{(N)} & \xrightarrow{\rho} & \mathbb{C} \\
 \Psi & & \Psi \\
 \zeta^{\mathfrak{m}} \left(\begin{smallmatrix} k_1, & \dots, & k_d \\ \epsilon_1, & \dots, & \epsilon_d \end{smallmatrix} \right) & \mapsto & \zeta \left(\begin{smallmatrix} k_1, & \dots, & k_d \\ \epsilon_1, & \dots, & \epsilon_d \end{smallmatrix} \right) \\
 \text{motivic CMZVs} & & \text{CMZVs}
 \end{array}$$

$\mathcal{H}^{(N)} = \bigoplus_{w=0}^{\infty} \mathcal{H}_w^{(N)}$: The graded ring defined by an algebraic way.

(More precisely, the ring of effective motivic periods of mixed Tate motives associated to $\mathbb{Q}(\mu_N)$ and $\langle 1 - a \mid a \in \mu_N \setminus \{1\} \rangle \otimes \mathbb{Q} \subset \mathbb{Q}(\mu_N)^{\times} \otimes \mathbb{Q}$.)

Let $Z_w^{(N), \mathfrak{m}} := \langle \text{motivic CMZVs of level } N \text{ and weight } w \rangle_{\mathbb{Q}} \subset \mathcal{H}_w^{(N)}$

* By definition $\dim_{\mathbb{Q}} Z_w^{(N)} \leq \dim_{\mathbb{Q}} Z_w^{(N), \mathfrak{m}}$.

* Conjecturally, ρ is injective and so $\dim_{\mathbb{Q}} Z_w^{(N)} = \dim_{\mathbb{Q}} Z_w^{(N), \mathfrak{m}}$.

Upper bound (1/2)

Let

$$\tilde{\mathcal{H}}^{(N)} := \begin{cases} (\text{real part of } \mathcal{H}^{(N)}) & N = 1, 2 \\ \mathcal{H}_N & N \geq 3 \end{cases} \quad (\supset Z^{(N), \mathfrak{m}})$$

Then the dimension of $\tilde{\mathcal{H}}_w^{(N)}$ is given by

$$\dim_{\mathbb{Q}} \tilde{\mathcal{H}}_w^{(N)} = d_w^{(N)}$$

where

$$\sum_{w=0}^{\infty} d_w^{(N)} t^w := \begin{cases} \frac{1}{1-t^2-t^3} & N = 1 \\ \frac{1}{1-t-t^2} & N = 2 \\ \frac{1}{1 - (\frac{\varphi(N)}{2} + \nu(N))t + (\nu(N) - 1)t^2} & N \geq 3 \end{cases}$$

(φ is Euler's totient function and $\nu(N)$ is number of prime divisors of N .)

Upper bound (2/2)

$Z^{(N),\mathfrak{m}} \subset \tilde{\mathcal{H}}_w^{(N)}$ and $\dim_{\mathbb{Q}} \tilde{\mathcal{H}}_w^{(N)} = d_w^{(N)}$ implies the following theorem.

Theorem (Deligne-Goncharov)

For $N \geq 1$ and $w \geq 0$, we have $\dim_{\mathbb{Q}} Z_w^{(N),\mathfrak{m}} \leq d_w^{(N)}$.

Corollary (Deligne-Goncharov)

For $N \geq 1$ and $w \geq 0$, we have $\dim_{\mathbb{Q}} Z_w^{(N)} \leq d_w^{(N)}$.

The above upper bound is not necessarily strict.

When $\dim_{\mathbb{Q}} Z_w^{(N),\mathfrak{m}} = d_w^{(N)}$?

Previous results

Theorem (Goncharov)

When $N \geq 5$ is a prime number,

$$\dim_{\mathbb{Q}} Z_2^{(N),\mathfrak{m}} < d_2^{(N)}.$$

Theorem (Deligne for $N = 2,3,4,8$; Brown for $N = 1$)

For $N \in \{1,2,3,4,8\}$, we have

$$\dim_{\mathbb{Q}} Z_w^{(N),\mathfrak{m}} = d_w^{(N)}.$$

- * The above theorem is equivalent to $Z_w^{(N),\mathfrak{m}} = \tilde{\mathcal{H}}_w^{(N)}$ for $N = \{1,2,3,4,8\}$.
- * They also give explicit basis for $Z_w^{(N),\mathfrak{m}}$ for $N = \{1,2,3,4,8\}$.

Main theorem and corollaries (1/2)

Theorem (H.)

For $N = p^r \geq 3$ with $p \in \{2,3\}$, a basis of $\mathcal{H}^{(N)}$ is given by

$$\left\{ (2\pi i)_{\mathfrak{m}}^s \zeta^{\mathfrak{m}} \left(\begin{matrix} k_1, \dots, k_d \\ \zeta_N^{a_1}, \dots, \zeta_N^{a_d} \end{matrix} \right) \middle| \begin{matrix} s, d \geq 0, k_1, \dots, k_d \geq 1, a_1, \dots, a_d \in \mathbb{Z}/N\mathbb{Z} \\ a_1 \equiv \dots \equiv a_{d-1} \equiv 0, a_d \equiv 1 \pmod{p^{4-p}} \end{matrix} \right\}.$$

Corollary (H.)

For $N = p^r \geq 3$ with $p \in \{2,3\}$,

$$\dim_{\mathbb{Q}} Z_w^{(N), \mathfrak{m}} = d_w^{(N)} \quad \left(= \left(\frac{p^{r-1}(p-1)}{2} + 1 \right)^w \right).$$

Main theorem and corollaries (2/2)

Corollary (H.)

For $N = p^r \geq 3$ with $p \in \{2,3\}$, all **motivic** CMZVs of level N can be **uniquely** written as a $\mathbb{Q}$ -linear sum of

$$\left\{ (2\pi i)_m^s \zeta^m \left(\begin{matrix} k_1, \dots, k_d \\ \zeta_N^{a_1}, \dots, \zeta_N^{a_d} \end{matrix} \right) \middle| \begin{matrix} s, d \geq 0, k_1, \dots, k_d \geq 1, a_1, \dots, a_d \in \mathbb{Z}/N\mathbb{Z} \\ a_1 \equiv \dots \equiv a_{d-1} \equiv 0, a_d \equiv 1 \pmod{p^{4-p}} \end{matrix} \right\}.$$

Corollary (H.)

For $N = p^r \geq 3$ with $p \in \{2,3\}$, all CMZVs of level N can be written as a $\mathbb{Q}$ -linear sum of

$$\left\{ (2\pi i)^s \zeta \left(\begin{matrix} k_1, \dots, k_d \\ \zeta_N^{a_1}, \dots, \zeta_N^{a_d} \end{matrix} \right) \middle| \begin{matrix} s, d \geq 0, k_1, \dots, k_d \geq 1, a_1, \dots, a_d \in \mathbb{Z}/N\mathbb{Z} \\ a_1 \equiv \dots \equiv a_{d-1} \equiv 0, a_d \equiv 1 \pmod{p^{4-p}} \end{matrix} \right\}.$$

Hopf structure and filtration

Hopf structure

Hereafter, we fix $N \geq 1$.

We omit (N) from the notation if there is no risk of confusion.

* Put $\mathcal{A} := \mathcal{H}/(2\pi i)_{\mathfrak{m}}$ and $\zeta^{\mathfrak{a}}(\dots) = (\zeta^{\mathfrak{m}}(\dots) \bmod (2\pi i)_{\mathfrak{m}}) \in \mathcal{A}$.

* It is known that $\mathcal{A}$ has a structure of Hopf algebra.
(We can calculate coproduct Δ by Goncharov's formula.)

* Define the coradical filtration

$$\{0\} = C_{-1}\mathcal{A} \subset C_0\mathcal{A} \subset C_1\mathcal{A} \subset C_2\mathcal{A} \subset \dots$$

$$\text{by } C_d\mathcal{A} = \{u \in \mathcal{A} \mid \Delta(u) - 1 \otimes u \in \mathcal{A} \otimes C_{d-1}\mathcal{A}\} \quad (d \geq 0).$$

* Put $\text{gr}_d^C\mathcal{A} := C_d\mathcal{A}/C_{d-1}\mathcal{A}$.

* Δ induces an isomorphism $\text{gr}_d^C\mathcal{A} \simeq \text{gr}_1^C\mathcal{A} \otimes \text{gr}_{d-1}^C\mathcal{A} \simeq \dots \simeq (\text{gr}_1^C\mathcal{A})^{d\otimes}$.

Coproduct formula - Iterated integral symbol

For $a_0, \dots, a_{k+1} \in \mu_N \cup \{0\}$, define $I^{\mathfrak{m}}(a_0; a_1, \dots, a_k; a_{k+1}) \in \mathcal{H}^{(N)}$ by the following rules:

$$(1) \quad I^{\mathfrak{m}}(a_0; a_1, \dots, a_k; a_{k+1}) := \sum_{j=0}^k (-1)^j I^{\mathfrak{m}}(0; a_j, \dots, a_1; a_0) \cdot I^{\mathfrak{m}}(0; a_{j+1}, \dots, a_k; a_{k+1})$$

$$(2) \quad I^{\mathfrak{m}}(0; a_1, \dots, a_k; 0) = \delta_{k,0}$$

$$(3) \quad \sum_{j=0}^k I^{\mathfrak{m}}(0; a_1, \dots, a_j, s, a_{j+1}, \dots, a_k; \epsilon) = 0 \quad (s \in \{0, \epsilon\})$$

$$(4) \quad I^{\mathfrak{m}}(0; z_1, \{0\}^{k_1-1}, \dots, z_d, \{0\}^{k_d-1}; z_{d+1}) := (-1)^d L^{\mathfrak{m}} \left(\begin{array}{ccc} \frac{z_2}{z_1}, & \dots, & \frac{z_{d+1}}{z_d} \\ k_1, & \dots, & k_d \end{array} \right) \quad (z_j \in \mu_N)$$

Coproduct formula

We put $I^a(a_0; a_1, \dots, a_k; a_{k+1}) := (I^m(a_0; a_1, \dots, a_k; a_{k+1}) \bmod (2\pi i)_m) \in \mathcal{A}^{(N)}$

Theorem (Goncharov)

$$\begin{aligned} & \Delta I^m(a_0; a_1, \dots, a_k; a_{k+1}) \\ &= \sum_{n=0}^k \sum_{0=i_0 < i_1 < \dots < i_n < i_{n+1}=a_{k+1}} \left(\prod_{j=0}^n I^a(a_{i_j}; a_{i_{j+1}}, \dots, a_{i_{j+1}-1}; a_{i_{j+1}}) \right) \otimes I^m(a_{i_0}; a_{i_1}, \dots, a_{i_n}; a_{i_{n+1}}) \end{aligned}$$

For example,

$$\begin{aligned} \Delta I^m(a; b, c; d) &= I^a(a; b, c; d) \otimes I^m(a; d) \\ &\quad + I^a(a; b; c) \cdot I^a(c; d) \otimes I^m(a; c; d) + I^a(a; b) \cdot I^a(b; c; d) \otimes I^m(a; b; d) \\ &\quad + I^a(a; b) \cdot I^a(b; c) \cdot I^a(c; d) \otimes I^m(a; b, c; d) \\ &= I^a(a; b, c; d) \otimes 1 \\ &\quad + I^a(a; b; c) \otimes I^m(a; c; d) + I^a(b; c; d) \otimes I^m(a; b; d) \\ &\quad + 1 \otimes I^m(a; b, c; d) \end{aligned}$$

Hopf structure

- * We can show $\zeta^{\mathbf{a}} \begin{pmatrix} k_1, & \dots, & k_d \\ \epsilon_1, & \dots, & \epsilon_d \end{pmatrix} \in C_d \mathcal{A}$.
- * The structure of $\text{gr}_1^C \mathcal{A}$ is well-understood.
 - $\text{gr}_1^C \mathcal{A}$ is spanned by $\zeta^{\mathbf{m}} \begin{pmatrix} k \\ \epsilon \end{pmatrix}$.
 - All the linear relations among $\zeta^{\mathbf{m}} \begin{pmatrix} k \\ \epsilon \end{pmatrix}$ is known.
- * Moreover, we can calculate the image of $\zeta^{\mathbf{a}} \begin{pmatrix} k_1, & \dots, & k_d \\ \epsilon_1, & \dots, & \epsilon_d \end{pmatrix}$
in $\text{gr}_d^C \mathcal{A} \simeq (\text{gr}_1^C \mathcal{A})^{d \otimes}$ explicitly.

Previous results

- * Let $P(N, w, d)$ be the statement that $\mathrm{gr}_d^C \mathcal{A}_w$ is spanned by the images of CMZVs of level N , weight w , depth d .
- * There is an explicit algorithm to determine $P(N, w, d)$

Theorem (Deligne for $N = 2, 3, 4, 8$; H. for general $N \geq 4$)

$P(N, w, d)$ holds when $N = p^r \geq 2$ with $p \in \{2, 3\}$.

- * This implies the main theorem.
- * $P(1, 12, 2)$ does not hold, which are closely related to the existence of nonzero weight 12 cusp forms for $SL(2, \mathbb{Z})$.

Proof of main theorem

Structure of $\mathcal{A}_1^{(N)}$

In this section, we assume $N \geq 2$.

For $a \in \mathbb{Z}/N\mathbb{Z}$, put $\langle a \rangle := I^{\mathfrak{m}}(0; \zeta_N^a; 1) = \zeta^{\mathfrak{m}} \begin{pmatrix} 1 \\ \zeta_N^{-a} \end{pmatrix}$.

Theorem (Deligne-Goncharov)

$\mathcal{A}_1^{(N)}$ is spanned by $[a]$ for $a \in \mathbb{Z}/N\mathbb{Z}$ with the following relations:

- * $\langle 0 \rangle = 0$,
- * $\langle a \rangle = \langle -a \rangle$ for $a \in \mathbb{Z}/N\mathbb{Z}$,
- * $\langle b \rangle = \sum_{aM=b} \langle a \rangle$ for $M \mid N$ and $b \neq 0 \in M\mathbb{Z}/N\mathbb{Z}$.

How to calculate (3/3)

Furthermore, the image of $\Delta I^a(0; \zeta_N^b, \zeta_N^a; 1)$ in $\mathcal{A}_1^{(N)} \otimes \mathcal{A}_1^{(N)}$ can be calculated by using Goncharov's coproduct formula as

$$\begin{aligned} & I^a(\zeta_N^a; \zeta_N^b; 1) \otimes I^a(0; \zeta_N^a; 1) + I^a(0; \zeta_N^a; \zeta_N^b) \otimes I^a(0; \zeta_N^b; 1) \\ &= I^a(0; \zeta_N^b; 1) \otimes I^a(0; \zeta_N^a; 1) - I^a(0; \zeta_N^{b-a}; 1) \otimes I^a(0; \zeta_N^a; 1) + I^a(0; \zeta_N^{b-a}; 1) \otimes I^a(0; \zeta_N^b; 1) \\ &= \langle b \rangle \otimes \langle a \rangle - \langle b - a \rangle \otimes \langle a \rangle + \langle b - a \rangle \otimes \langle b \rangle. \end{aligned}$$

Thus $\mathcal{H}_2^{(N)}$ is spanned by motivic CMZVs of level N , weight 2 if and only if the following map is surjective:

$$\begin{array}{ccc} \tilde{\Delta} : & \mathbb{Q}[\mathbb{Z}/N\mathbb{Z}] \otimes \mathbb{Q}[\mathbb{Z}/N\mathbb{Z}] & \longrightarrow & \mathcal{A}_1^{(N)} \otimes \mathcal{A}_1^{(N)} \\ & \Downarrow & & \Downarrow \\ & [a] \otimes [b] & \longmapsto & \langle a \rangle \otimes \langle b \rangle - \langle a \rangle \otimes \langle b - a \rangle + \langle b \rangle \otimes \langle b - a \rangle. \end{array}$$

Combinatorial description for weight 2

- Furthermore, let $N = p^r \geq 3$ with $p \in \{2,3\}$, $q = p^{2-p}$.

- Put $W := \bigoplus_{\substack{a \in \mathbb{Z}/N\mathbb{Z} \\ a \equiv 1 \pmod{q}}} \mathbb{Q}[a] \subset \mathbb{Q}[\mathbb{Z}/N\mathbb{Z}]$.

- Since W corresponds the given basis in main theorem,
what we want to show is the surjectivity of $W \otimes W \xrightarrow{\tilde{\Delta}} \mathcal{A}_1^{(N)} \otimes \mathcal{A}_1^{(N)}$.

Proof of main theorem for weight 2 case (1/6)

- * There is a bijection $W \simeq \mathcal{A}_1^{(N)}$; $[a] \mapsto \langle a \rangle$.

For example, when $N = 8$,

$W = \mathbb{Q}[1] + \mathbb{Q}[5]$ and

$$\langle 2 \rangle = \langle 1 \rangle + \langle 5 \rangle$$

$$\langle 3 \rangle = \langle 5 \rangle$$

$$\langle 4 \rangle = 2\langle 1 \rangle + 2\langle 5 \rangle$$

$$\langle 6 \rangle = \langle 1 \rangle + \langle 5 \rangle$$

$$\langle 7 \rangle = \langle 1 \rangle$$

$$\langle 0 \rangle = 0.$$

- * Thus, main theorem is equivalent to the bijectivity of

$$\rho : W \otimes W \xrightarrow{\tilde{\Delta}} \mathcal{A}_1^{(N)} \otimes \mathcal{A}_1^{(N)} \simeq W \otimes W.$$

Proof of main theorem for weight 2 case (2/6)

Let $\bar{\rho}$ be a modulo p reduction of ρ .

Main theorem

$\Leftrightarrow \rho$ is bijective

$\Leftrightarrow \det(\rho) \neq 0$

$\Leftarrow \det(\rho) \not\equiv 0 \pmod{p}$

$\Leftrightarrow \bar{\rho}$ is bijective

$\Leftarrow \bar{\rho}$ is unipotent

Thus main theorem is proved if the unipotency of $\bar{\rho}$ is proved.

Proof of main theorem for weight 2 case (3/6)

Let us see some examples of $\bar{\rho}$ by standard basis.

For $N = 4$, $\bar{\rho} = (1)$.

$$\text{For } N = 8, \bar{\rho} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

For $N = 16$,

[illegible]

Proof of main theorem for weight 2 case (4/6)

Thus, when $N = 16$, $\bar{\rho}$ is not identity, but

$$(\bar{\rho} - 1)^2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(\bar{\rho} - 1)^3 = 0$$

Proof of main theorem for weight 2 case (5/6)

- * Let $W_p := \bigoplus_{\substack{a \in \mathbb{Z}/N\mathbb{Z} \\ a \equiv 1 \pmod{q}}} \mathbb{F}_p[a] \subset \mathbb{F}_p[\mathbb{Z}/N\mathbb{Z}]$ be a mod p reduction of W .
- * Define a subgroup G of $(\mathbb{Z}/N\mathbb{Z})^\times$ by $G = \{x \equiv 1 \pmod{q}\}$
- * Define the action of G to $W_p \otimes W_p$ by $\sigma_x([a] \otimes [b]) = [xa] \otimes [xb]$.
- * $I_G := \ker(\mathbb{F}_p[G] \rightarrow \mathbb{F}_p)$.
- * For $m, n \geq 0$, define $V(m, n) \subset W_p \otimes W_p$ as follows:
 - * When $n \leq r$ with $N = qp^r$, let $V(m, n)$ be the subspace spanned by
$$\sum_{i=0}^{p^n} g([a] \otimes [b + iqp^{r-n}]) \quad \text{where } g \in I_G^m, a, b \in \mathbb{Z}/N\mathbb{Z}, a \equiv b \equiv 1 \pmod{q}$$
 - * When $n > r$, let $V(m, n) = \{0\}$.

Proof of main theorem for weight 2 case (6/6)

Then, we can show

$$(\bar{\rho} - \text{id})V(m, n) \subset V(m, n + 1) + V(m + 1, 1).$$

For example, let $N = 27$ and $x = [1] \otimes ([4] + [13] + [22]) \in V(0, 1)$.

Then $(\bar{\rho} - \text{id})(x) = -y_1 + y_2$ where

$$y_1 = [1] \otimes ([1] + [4] + [7] + [10] + \dots + [25]) \in V(0, 2)$$

$$\begin{aligned} y_2 &= + [4] \otimes ([1] + [10] + [19]) \\ &\quad + [13] \otimes ([4] + [13] + [22]) \\ &\quad + [22] \otimes ([7] + [16] + [25]) \\ &= + \sigma_4([1] \otimes ([7] + [16] + [25])) \\ &\quad + \sigma_{13}([1] \otimes ([1] + [10] + [19])) \\ &\quad + \sigma_{22}([1] \otimes ([4] + [13] + [22])) \\ &= + (\sigma_4 - 1)([1] \otimes ([7] + [16] + [25])) \\ &\quad + (\sigma_{13} - 1)([1] \otimes ([1] + [10] + [19])) \\ &\quad + (\sigma_{22} - 1)([1] \otimes ([4] + [13] + [22])) \\ &\quad + [1] \otimes ([1] + [4] + [7] + \dots + [22]) \\ &\in V(1, 1) + V(0, 2). \end{aligned}$$

Proof of main theorem for weight 2 case (6/6)

- * We can show

$$(\bar{\rho} - \text{id})V(m, n) \subset V(m, n + 1) + V(m + 1, 0).$$

Note that $(m, n + 1)$ and $(m + 1, 0)$ are greater than (m, n) in lexicographic order.

- * $V(m, n) = \{0\}$ when $m \geq p^r + 1$ or $n \geq r + 1$ since $I_G^{p^r} = \{0\}$.
- * Thus, $\bar{\rho} - \text{id}$ is nilpotent, which completes the proof of main theorem for weight 2 case.
- * The proof for general weight case is more complicated, but similar.

Some observations for
other level, weight 2 case

Combinatorial description for weight 2

$\mathcal{H}_2^{(N)}$ is spanned by motivic CMZVs of level N , weight 2

if and only if the following map is surjective:

$$\begin{array}{ccc} \tilde{\Delta} : & \mathbb{Q}[\mathbb{Z}/N\mathbb{Z}] \otimes \mathbb{Q}[\mathbb{Z}/N\mathbb{Z}] & \longrightarrow \mathcal{A}_1^{(N)} \otimes \mathcal{A}_1^{(N)} \\ & \Downarrow & \Downarrow \\ & [a] \otimes [b] & \longmapsto \langle a \rangle \otimes \langle b \rangle - \langle a \rangle \otimes \langle b - a \rangle + \langle b \rangle \otimes \langle b - a \rangle. \end{array}$$

- * Let D_N be the dimension of $\text{coker} \tilde{\Delta}$.
- * Main theorem implies that $D_{2r} = D_{3r} = 0$.
- * Goncharov shows that $D_p = \frac{p^2 - 1}{24}$ for a prime $p \geq 5$.
- * It seems those are all known formulas for exact values of D_N .

The case $N = 2p$ or $3p$ for prime p

What are D_{2p} and D_{3p} ?

Is it similar to D_p ?

p	5	7	11	13	17	19	23	29	31	37	41	43	...	83	...	113
D_p	1	2	5	7	12	15	22	35	40	57	70	77	...	330	...	532
D_{2p}	0	0	0	0	1	0	0	0	2	0	1	2	...	0	...	3
D_{3p}	0	0	0	1	0	0	0	0	0	1	4	0	...	0		

Conjecture (H.-Sato)

Let $f(p)$ (resp. $g(p)$) be the order of 2 (resp. 3) in $(\mathbb{Z}/p\mathbb{Z})^\times / \{\pm 1\}$.

$$\text{Then } D_{2p} = \frac{p-1}{2f(p)} - 1 \quad \text{and} \quad D_{3p} = \frac{p-1}{2g(p)} - 1.$$

Theorem (H.)

$$\text{Under the same settings, } D_{2p} \geq \frac{p-1}{2f(p)} - 1 \quad \text{and} \quad D_{3p} \geq \frac{p-1}{2g(p)} - 1.$$

The case $N = 5p$ for prime p

p	7	11	13	17	19	23	29	31	37	41	43
D_{5p}	0	2	8	8	1	0	16	6	18	23	0

p	47	53	59	61	67	71	73	79	83	89
D_{5p}	46	26	1	94	2	8	36	1	0	317

I have not found any pattern yet.

Other observations

For $N \leq 400$, the following is true.

✱ $D_{p^2} = \binom{p}{4} = \frac{p(p-1)(p-2)(p-3)}{24}$ for a prime p .

(This was already conjectured by J. Zhao).

✱ For $a, b \geq 1$ and primes p, q , $D_{p^a q^b} = D_{pq}$.

Integral structure

Conjecture about the integral structure

Put $\mathbb{Z}_{(p)} = \{a/b : b \not\equiv 0 \pmod{p}\}$

Conjecture

For $N = p^r \geq 3$ with $p \in \{2,3\}$, all CMZVs of level N can be written as a $\mathbb{Z}_{(p)}$ -linear sum of

$$\left\{ \frac{(2\pi i/N)^s}{s!} \zeta \left(\begin{matrix} k_1, \dots, k_d \\ \zeta_N^{a_1}, \dots, \zeta_N^{a_d} \end{matrix} \right) \mid s, d \geq 0, k_1, \dots, k_d \geq 1, a_1, \dots, a_d \in \mathbb{Z}/N\mathbb{Z} \right. \\ \left. a_1 \equiv \dots \equiv a_{d-1} \equiv 0, a_d \equiv 1 \pmod{p^{4-p}} \right\}.$$

Conjecture

All MZVs can be written as a $\mathbb{Z}_{(2)}$ -linear sum of $\{\zeta(k_1, \dots, k_d) \mid k_1, \dots, k_d \in \{2,3\}\}$.

Thank you for listening

Appendix - Table of D_N for $N \leq 250$

N	3	4	5	6	7	8	9	10	11	12	13	14
D_N	0	0	1	0	2	0	0	0	5	0	7	0

N	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
D_N	0	0	12	0	15	0	0	0	22	0	5	0	0	0	35

N	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44
D_N	0	40	0	0	1	0	0	57	0	1	0	70	0	77	0

N	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59
D_N	0	0	92	0	35	0	0	0	117	0	2	0	0	0	145

N	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74
D_N	0	155	2	0	0	8	0	187	1	0	0	210	0	222	0

N	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
D_N	0	0	15	0	260	0	0	1	287	0	8	2	0	0	330

Appendix - Table of D_N for $N \leq 250$

N	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104
D_N	0	32	0	0	0	1	0	392	0	0	0	425	0	442	0

N	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119
D_N	0	0	477	0	495	0	1	0	532	0	0	0	1	0	72

N	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134
D_N	0	330	0	4	2	25	0	672	0	0	0	715	0	99	0

N	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149
D_N	0	1	782	0	805	0	0	0	240	0	16	3	0	0	925

N	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164
D_N	0	950	0	0	0	6	0	1027	0	0	0	165	0	1107	1

N	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179
D_N	0	0	1162	0	715	0	0	2	1247	0	0	0	0	3	1335

Appendix - Table of D_N for $N \leq 250$

N	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194
D_N	0	1365	0	5	0	18	0	440	0	0	0	1520	0	1552	1

N	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209
D_N	0	0	1617	0	1650	0	2	0	294	0	23	0	0	0	585

N	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224
D_N	0	1855	0	0	0	0	0	345	2	5	0	720	0	2072	0

N	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239
D_N	0	3	2147	0	2185	0	0	0	2262	0	46	0	0	0	2380

N	240	241	242	243	244	245	246	247	248	249	250
D_N	0	2420	0	0	0	0	0	954	2	0	0